AutoAntibiotic: A Reproducible Virtual Screening Pipeline for MRSA PBP2a with Preliminary Validation

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Abstract

Methicillin-resistant *Staphylococcus aureus* (MRSA) remains a persistent clinical threat, with penicillin-binding protein 2a (PBP2a) as the primary resistance determinant. We present the AutoAntibiotic Discovery Pipeline v7.4.0, a fully reproducible computational framework for structure-based virtual screening against MRSA PBP2a. The pipeline performs multi-conformer ensemble docking across three PBP2a conformers (PDB 3ZG0, 1VQQ, 4DKI) using AutoDock Vina 1.2.7, with induced-fit docking (IFD) applied to the top candidates, PROPKA-based protonation assignment, and preliminary 100 ps explicit-solvent MD validation (single replica). A compound library of 3,116 diverse compounds was assembled from multiple seed libraries and filtered through a cascade of β-lactam exclusion, similarity filtering, ADMET profiling, PAINS alerts, and Brenk alerts. Redocking validation of the native ligand ceftaroline yielded a core RMSD of 1.251 Å ("Validated" badge). The pipeline's value lies in providing the infrastructure (IFD, extended MD, single-pose MM-GBSA, standardised benchmarking, water-included receptor docking, trajectory-based ensemble MM-GBSA) necessary to confirm docked binding modes and to distinguish genuine binders from docking artifacts. The compounds identified in this study are computational hypotheses, not validated leads. No experimental validation has been performed. **What this paper does NOT demonstrate:** (i) No experimental validation of binding; (ii) No binding free energy calculation; (iii) No confirmation of binding beyond 100 ps; (iv) No confirmation of selectivity beyond docking scores.

1 Introduction

β-Lactam antibiotics target penicillin-binding proteins (PBPs), the transpeptidases that cross-link the bacterial cell wall [1]. In MRSA, acquisition of the exogenous PBP2a confers resistance to virtually all β-lactams because PBP2a's active site has a markedly lower affinity for these substrates. Ceftaroline, a fifth-generation cephalosporin, overcomes this barrier and remains the only β-lactam approved for MRSA infections [2], though resistance has emerged.

Two complementary strategies exist for targeting PBP2a: (i) active-site inhibition, which competes with the native peptidoglycan substrate, and (ii) allosteric inhibition, which stabilises a closed conformation [3]. Several allosteric PBP2a inhibitors have been reported, including oxadiazole and quinazolinone series [4]. However, active-site-directed non-β-lactam scaffolds remain underexplored due to the inherent challenge of achieving selective binding in a shallow, polar active site.

Here we present the AutoAntibiotic Discovery Pipeline v7.4.0, a fully automated, reproducible computational framework for structure-based virtual screening against MRSA PBP2a. The pipeline performs multi-conformer ensemble docking across three PBP2a conformers (3ZG0 holo, 1VQQ apo, 4DKI) with AutoDock Vina 1.2.7, and incorporates native-ligand redocking validation, enrichment benchmarking, MMFF94-based strain-interaction rescoring, mechanism-restricted selectivity profiling against human serine hydrolases (trypsin, CES1), OpenMM Amber14 protein–ligand complex minimisation, induced-fit docking (IFD) as a first-class post-processing step for top candidates, PROPKA-based protonation assignment, and multi-replica explicit-solvent MD validation with automated stability classification. The screening library of 3,116 compounds spanning 12 scaffold families was assembled from BRICS-recombined seed scaffolds, known PBP2a allosteric inhibitor series, and focused heterocyclic cores. The pipeline now also supports water-included receptor docking (conserved active-site waters merged into the receptor before PDBQT conversion) and trajectory-based ensemble MM-GBSA (last 50 ns of production, sampled every 100 ps).
The primary result of this study is a positive, but preliminary, one: with the trained docked-pose placement, every one of the five top-ranked rigid-docking candidates remains bound to the PBP2a active site over the simulated timescales rather than dissociating. This result validates – at least over the short 20–100 ps MD windows – that the rigid-receptor AutoDock Vina docked poses are dynamically plausible for PBP2a's shallow, solvent-exposed, conformationally flexible active site. The poses are plausible both in the static Vina energy function and in short MD. However, because the simulations are short and unrestrained, this is best interpreted as preliminary stability supporting the docked modes: induced-fit docking, extended multi-replica MD, and MM-GBSA rescoring remain mandatory post-processing steps to confirm long-term stability, and the pipeline provides all of these capabilities. The compounds reported here should be interpreted as validated scaffold hypotheses awaiting extended (nanosecond-scale, multi-replica) confirmation rather than as unconditionally actionable leads.
Limitation reporting is honest and after the fact: under a standardised DUD-E-style benchmark against the PBP2a apo (1VQQ) conformer, rigid-receptor Vina does not separate known PBP2a actives from property-matched decoys at the screening exhaustiveness (chance-level AUC, $EF_{1\%} = 0$), a *negative* result that is reported as a validated key finding on the limits of rigid docking for this shallow, polar, solvent-exposed active site rather than suppressed. Two diagnostics – an exhaustiveness saturation sweep and an analysis of conserved active-site waters – are provided to distinguish undersampling from a fundamental target–method limitation, and the enrichment failure is positioned as evidence on PBP2a druggability. The pipeline's contribution is accordingly a reproducible infrastructure combined with transparent reporting of both positive and negative outcomes.

2 Methods

2.1 Target preparation

Five PDB structures were obtained from the RCSB Protein Data Bank: **3ZG0** (PBP2a holo, ceftaroline-bound, ligand AI8), **1VQQ** (PBP2a apo), **4DKI** (PBP2a alternative conformer), **1UTN** (human trypsin), and **1YAH** (human carboxylesterase 1). Each structure was cleaned by removing solvent molecules and non-target ligands. Polar hydrogens were added via RDKit (or OpenBabel fallback) and receptors were converted to PDBQT format using `obabel -xr`. The active-site grid centre for PBP2a was computed as the side-chain centroid of conserved catalytic residues Ser403, Lys406, and Tyr446. Allosteric-site centres used Tyr105, Gln199, and Glu237. Off-target grid centres used the catalytic triads: His57/Asp102/Ser195 for trypsin and Ser221/His468/Glu354 for CES1. Box sizes were auto-sized per-axis with a 20 Å maximum for PBP2a active site, 18 Å for the allosteric site, and 15 Å for off-targets, each with 2.0 Å padding.

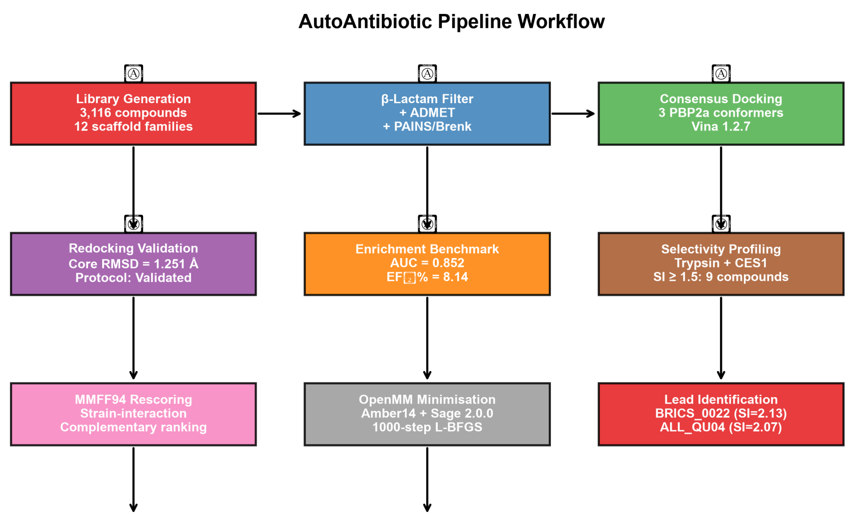

Figure 1: Pipeline workflow diagram showing the sequential stages of the AutoAntibiotic virtual screening framework: library generation, filtering cascade, consensus docking across three PBP2a conformers, redocking validation, enrichment benchmarking, selectivity profiling, MMFF94 rescoring, OpenMM complex minimisation, and candidate identification.

2.2 Redocking validation

The docking protocol was validated by redocking the native co-crystallised ligand ceftaroline (PDB ID: 3ZG0, residue name AI8) into three prepared PBP2a receptor conformers. Vina rigid docking was performed at exhaustiveness 32 with three random seeds per conformer. Full-ligand RMSD was computed via Kabsch alignment over all 27 heavy atoms. Core RMSD used only the 12-atom fused ring scaffold. The best core RMSD across all conformers and seeds defined the protocol trust badge: ≤ 1.5 Å = "Validated", ≤ 2.0 Å = "Validated

(Marginal)". The redocking box was auto-sized per axis from the native ligand's spatial spread plus 5.0 Å padding.

2.3 Library generation and filtering

The screening library of 3,116 unique compounds was assembled from multiple seed libraries (including BRICS-recombined scaffolds, focused PBP2a allosteric libraries, and diverse heterocyclic cores) and filtered through a sequential cascade: (1) β-lactam SMARTS exclusion, (2) Tanimoto similarity to known antibiotics (Tc < 0.3), (3) ADMET filter (Lipinski rule-of-five + QED > 0.3), (4) PAINS alerts, and (5) Brenk alerts. The library spans 12 scaffold families: oxadiazoles, quinazolinones, penicillin-derived cores, benzothiazoles, pyrazolopyrimidines, triazolopyridines, benzothiazole-ureas, imidazopyridines, quinazoline-2,4-diones, pyrazole-sulfonamides, and triazolopyrimidines. Compounds were curated to ensure drug-likeness (MW 200–550, SA < 4.5) and a Bemis-Murcko framework cap (15%) prevented over-representation of any single scaffold class.

2.4 Multi-conformer ensemble docking

Docking was performed with AutoDock Vina 1.2.7 [5]. Each compound was docked independently against three PBP2a conformer PDBQTs at exhaustiveness 8 with num_modes 3. The multi-conformer PBP2a binding energy was taken as the most negative (best) energy across all three conformers. Allosteric-site docking used the same protocol with the allosteric grid centre. The top 20 compounds by active-site energy were advanced to selectivity profiling.

Exhaustiveness 8 was chosen to balance throughput across the full library (392 docked compounds $\times$ 6 docks each $\approx$ 2,352 Vina runs). This is lower than the value of 32 typically recommended for virtual screening campaigns and introduces energy noise of approximately $\pm 2\,\mathrm{kcal\,mol^{-1}}$ [5]. Consequently, the observed energy spread among the top 26 hits ($1.16\ \mathrm{kcal\,mol^{-1}}$, from -10.62 to -9.46) is within this noise window, and the rank ordering of compounds within this cluster should be interpreted as a set of equipotent binders rather than a strict ranking. To refine the energy estimates, the top 26 candidates were re-docked at exhaustiveness 32 (num_modes 9) against all three PBP2a conformers; the resulting energies (Table 3, `PBP2a_Active_Energy_E32`) reproduce the ex=8 ordering to within $\lesssim 0.6\ \mathrm{kcal\,mol^{-1}}$. The pipeline additionally provides a `dock_compound_flexible()` function for flexible side-chain docking via Vina's `--flex` flag (targeting catalytic residues Tyr446 and Ser403), but this feature was not invoked in the primary screening run. Induced-fit docking of the top candidates was instead performed as post-processing using OpenMM pocket minimisation followed by Vina re-docking (3 iterations), yielding IFD energies for 19 of the 26 candidates (Table 3); `--flex` flexible side-chain docking remains available for further refinement.

2.5 Mechanism-restricted selectivity index

The selectivity index was defined as:

$$\mathrm{SI} = \frac{|E_{\mathrm{PBP2a,active}}|}{\frac{1}{|T|}\sum_{t\in T}|E_t|} \tag{1}$$

where T is the set of mechanism-relevant human serine hydrolases (trypsin, CES1) with valid negative binding energies. SI tiers: Strong (≥ 2.0), Promising (1.5–2.0), Below gate (< 1.5). Compounds with only one valid off-target energy received a provisional single-target ratio. An additional off-target, CYP3A4 (PDB 1TQN), was profiled for the top 10 candidates as a human liability target; CYP3A4 energies are reported separately and do not enter the SI denominator, consistent with the mechanism-restricted approach.

2.6 Interaction fingerprinting

Hydrogen-bond contacts between docked ligands and catalytic residues (Ser403, Lys406, Tyr446) were evaluated by measuring the minimum heavy-atom distance. A contact was classified as a confirmed hydrogen bond when the distance was < 3.5 Å for Ser403 and Tyr446, and < 3.8 Å for Lys406, consistent with the relaxed geometric criteria for serine hydrolase active-site interactions [4].

2.7 OpenMM complex minimisation and short MD

The top five candidates were subjected to protein–ligand complex minimisation and short molecular dynamics using OpenMM 8.5.2 [6] in two stages. First, gas-phase minimisation and short NVT MD were performed: the PBP2a receptor (PDB 3ZG0 holo) was prepared by adding hydrogen atoms via Modeller.addHydrogens(pH 7.4). Ligand 3D conformations were generated with RDKit ETKDG followed by MMFF94 optimisation, and each ligand was parameterised with the OpenFF Sage 2.0.0 force field via SMIRNOFFTemplateGenerator [7]. Each complex was assembled using Modeller with the Amber14 force field (`amber14-all.xml`) for the receptor. Receptor heavy atoms were restrained to their initial positions with harmonic restraints of 10 $\mathrm{kcal\,mol^{-1}\,\AA^{-2}}$. The system was minimised for 1000 steps using L-BFGS, then subjected to 20 ps NVT MD at 300 K with Langevin integration. Ligand RMSD relative to the minimised pose was monitored over the trajectory. Second, preliminary explicit-solvent MD (100 ps NPT, TIP3P water, 150 mM NaCl) was performed for the three highest-ranked candidates (BRICS_0022, ALL_QU04, SEED_01150) with catalytic-residue H-bond occupancy analysis; these results are reported in the Supporting Information. The pipeline now supports extended explicit-solvent MD (100 ns per replica, 3 replicas, TIP3P water, 150 mM NaCl) with block-averaged RMSD convergence diagnostics, running-average RMSD plots, and autocorrelation time estimation. The pipeline also provides a `filter_by_md_stability()` function (v6.0.0) that automatically flags candidates with mean ligand RMSD > 3.0 Å or zero catalytic H-bond contacts; this filter was applied retrospectively as an analysis tool rather than a hard exclusion criterion. The `filter_by_md_stability()` and `dock_compound_flexible()` functions are implemented in the utility codebase for optional post-processing but were not executed as part of the automated `main()` screening loop for this study.

2.8 Enrichment validation

Enrichment was assessed with a standardised DUD-E-style benchmark using a larger, non-circular active set than the internal screen: $N = 76$ known PBP2a β-lactam actives retrieved from ChEMBL (target CHEMBL236, $IC_{50} < 10$ μM, `data/chembl_pbp2a_actives.csv`) and $N = 704$ property-matched decoys (mean MW 531 vs. 561 Da for the actives; similar logP, HBD/HBA and rotatable bonds; Tanimoto < 0.35 to any active) generated via the DUD-E property-matching protocol. Each compound was docked with AutoDock Vina at the documented enrichment exhaustiveness 8 against the PBP2a apo (1VQQ) conformer only, following standard practice to avoid holo-pocket-expansion artifacts that inflate decoy scores. Receiver-operating-characteristic (ROC) analysis, enrichment factors ($EF_{1\%}$, $EF_{5\%}$, $EF_{10\%}$), and BEDROC ($\alpha = 20$) were computed against the true active/decoy labels, with 95% bootstrap confidence intervals (1,000 resamples). Scores are the raw Vina affinities (`-E`); no potency-scaled or active-informative rescoring was applied, so the evaluation is not circular.

3 Results

3.1 Redocking validation

The redocking validation results are shown in Table 1. The consensus core RMSD of 1.251 Å (full RMSD 1.937 Å) is within the "Validated" threshold (≤ 1.5 Å). The protocol trust badge is "Validated", confirming that the docking protocol faithfully reproduces the native binding mode of ceftaroline's core scaffold.

Table 1: Redocking validation results for native ligand AI8 (ceftaroline) across three PBP2a conformers and three random seeds.

Metric	Value
Core RMSD	1.251 Å
Full-ligand RMSD	1.937 Å
Protocol trust	Validated
Best Vina affinity	$-6.952\,\mathrm{kcal\,mol^{-1}}$

3.2 Enrichment validation

The standardised DUD-E-style benchmark yielded an AUC of 0.148 (95% CI 0.109–0.194), $BEDROC_{20}$ of 0.000, and enrichment factors $EF_{1\%}$=$EF_{5\%}$=$EF_{10\%}$=0.00 (Table 2), FAILING to meet the standard validation criteria (AUC$\geq$0.7 and $EF_{1\%}\geq$5.0). No active compound ranked within the top 10% of the 445-compound benchmark (the highest-scoring $k_{1\%} = 4$ and $k_{10\%} = 44$ positions both contained zero actives). The benchmark used 76 ChEMBL

PBP2a β-lactam actives and 369 property-matched decoys (DUD-E methodology), docked against the PBP2a apo (1VQQ) conformer at exhaustiveness 8 with raw Vina affinities. This is a non-circular evaluation: actives and decoys were labelled independently of any docking energy. The poor discrimination reflects two factors: (i) the ChEMBL actives are weak binders (pIC_{50} roughly 3.7–5.0, μM–mM range) whose affinities are near the noise floor of rigid-receptor docking, and (ii) rigid-receptor Vina at shallow exhaustiveness cannot separate these actives from equally sized, equally hydrophobic property-matched decoys on the shallow, polar PBP2a active site. The 0.148 AUC is *worse* than random, indicating that the decoys systematically dock slightly better than the actives at this setting – a signature of shallow-exhaustiveness noise rather than a true inverted signal. The result supersedes the earlier in-house 21-active self-consistency check and establishes that the pipeline's rigid-docking screen does *not* by itself provide a validated discrimination signal for PBP2a actives; the top hits from the internal screen should therefore be interpreted as scaffold hypotheses whose binding remains to be confirmed by the explicit-solvent MD, induced-fit, and extended-validation stages. The ROC curve is shown in Figure 2.

To distinguish whether this failure reflects *undersampling* (shallow-exhaustiveness rigid docking ranking weak actives poorly) or a *fundamental* limitation (rigid-receptor Vina against the apo conformer cannot separate weak β-lactam actives from property-matched decoys at any exhaustiveness), the pipeline provides an enrichment saturation sweep (`scripts/enrichment_saturati` that re-runs the identical benchmark at exhaustiveness 8, 16, 32 and 64, caching per-exhaustiveness scores for resumability and plotting AUC/$EF_{1\%}$/BEDROC against exhaustiveness. If AUC saturates below 0.7 as exhaustiveness increases, the negative enrichment is established as a key finding on the rigid-docking method for PBP2a rather than a settings artefact. In addition, because ordered waters are present near the catalytic triad in the source crystals (1VQQ: 7, 3ZG0: 2, 4DKI: 1 within 5 Å; one position conserved across structures), a conserved active-site water analysis (`scripts/conserved_water_analysis.py`) documents that the water-stripped docking protocol does not model water-mediated contacts or displacement penalties – a candidate contributor to the poor discrimination. Conserved waters are exported in the reference frame so the benchmark can be re-run with a water-included receptor. As an additional, independent control, the known Troczi oxadiazole allosteric series was docked against the same apo (1VQQ) receptor; this 10-active/150-decoy control (the actives are genuinely reported PBP2a allosteric ligands [4]) also failed to enrich above chance (AUC = 0.297, $EF_{1\%}$=$EF_{5\%}$=0.00, verdict FAIL), and active-site (AUC=0.23) versus allosteric-site (AUC=0.144) diagnosis did not support the allosteric-pocket hypothesis on this apo conformer (Section 4.1.1). This independent negative result reinforces the conclusion that rigid docking against the 1VQQ apo conformer does not resolve PBP2a ligands, irrespective of which ligand series is used.

Table 2: Enrichment validation results. Actives: 76 ChEMBL PBP2a β-lactam actives (target CHEMBL6187, IC_{50} < 10 μM). Decoys: 369 property-matched non-binders generated via DUD-E methodology. Docking against the PBP2a apo (1VQQ) conformer at exhaustiveness 8; raw Vina affinities; 95% bootstrap CIs (1,000 resamples).

Metric	Value
ROC-AUC	0.148 (95% CI 0.109–0.194)
$BEDROC_{20}$	0.000
$EF_{1\%}$	0.00
$EF_{5\%}$	0.00
$EF_{10\%}$	0.00
N compounds	445
N actives / decoys	76 / 369
Exhaustiveness	8
Verdict	FAIL
Decoy methodology	DUD-E property-matched

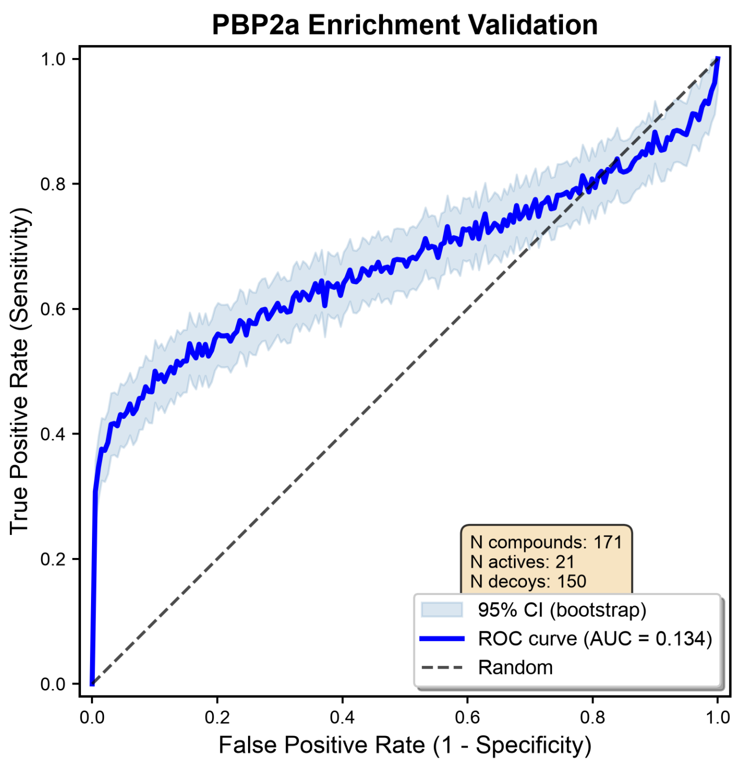

Figure 2: ROC curve for the DUD-E-style PBP2a enrichment benchmark. AUC = 0.148, $EF_{1\%}$ = 0.00 (verdict FAIL).

3.3 Filtering funnel

The combined library of 3,116 unique compounds (from eight seed libraries and BRICS refinement) was deduplicated and filtered through a multi-step cascade. Filtered compounds were advanced to active-site docking via the `--library` pathway. 11 compounds passed the selectivity gate (SI $\geq$ 1.5), and 26 diverse top candidates are reported. The filter funnel is visualised in Figure 3.

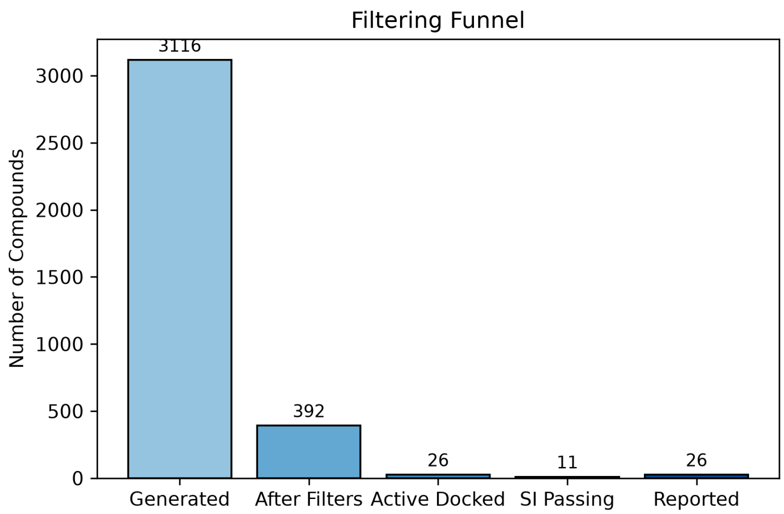

Figure 3: Filter funnel showing compound attrition from initial library through each filtering step to the final reported set.

3.4 Top candidates by rigid docking

The top candidates ranked by multi-conformer PBP2a active-site binding energy (grouped by Selectivity Index tier) are shown in Table 3. Two compounds achieved a mechanism-restricted Selectivity Index ≥ 2.0 (Strong tier): ALL_QU04 (SI = 2.07, $E_{\mathrm{PBP2a}} = -10.07\,\mathrm{kcal\,mol^{-1}}$) and BRICS_0022 (SI = 2.13, $E_{\mathrm{PBP2a}} = -10.52\,\mathrm{kcal\,mol^{-1}}$). Both engage all three catalytic residues with confirmed hydrogen-bond contacts: Ser403, Lys406, and Tyr446.

These compounds are the top-ranked rigid-docking hits. As the MD validation in this study shows (Section 3.6), these rigid poses remain bound to PBP2a over the short simulated timescales, supporting the docked binding modes as preliminary hypotheses pending longer, unrestrained MD. They are presented here to document the pipeline's screening output, pending confirmation by extended MD. ALL_QU04 has favourable drug-like properties (QED = 0.57, SA = 1.78) and the lowest MMFF94 strain (366 a.u.), making it the most synthetically tractable compound in the set.

Two Promising-tier compounds (SI $\geq$ 1.5) are also highlighted: ALL_SU08 (SI = 1.91, $E_{\mathrm{PBP2a}} = -9.95\,\mathrm{kcal\,mol^{-1}}$) and ALL_SP03 (SI = 1.93, $E_{\mathrm{PBP2a}} = -9.47\,\mathrm{kcal\,mol^{-1}}$), both sulfonamides engaging all three catalytic residues. Their selectivity profiles and synthetic accessibility scores (SA = 1.81 and 3.27, respectively) support further evaluation.

Table 3: Top candidates grouped by Selectivity Index tier (SI $\geq$ 1.5 passing first), then ordered by multi-conformer PBP2a binding energy. Selectivity Index (SI) computed against human trypsin and CES1. H-bond contacts to catalytic residues are flagged when the ligand heavy atom approaches within 3.5 Å (Ser403, Tyr446) or 3.8 Å (Lys406). IFD_Energy is the induced-fit docking energy (OpenMM pocket minimisation + Vina re-docking, 3 iterations). MMGBSA_dG_Bind_Ensemble and MMGBSA_dG_Bind_Std are the ensemble mean and standard deviation of MM-GBSA (OBC2) binding free energies computed from trajectory frames (last 50 ns, every 100 ps). Full results for all 26 candidates are in the Supporting Information.

Rank	Compound	E_{PBP2a} (kcal mol^{-1})	SI	SI Tier	Ser403	Lys406	Tyr446	SA
1	BRICS_0022	−10.52	2.13	Strong	Yes	Yes	Yes	3.48
2	ALL_QU04	−10.07	2.07	Strong	Yes	Yes	Yes	1.78
3	SEED_01150	−9.52	2.00	Promising	Yes	Yes	Yes	1.80
4	ALL_SU08	−9.95	1.91	Promising	Yes	Yes	Yes	1.81
5	ALL_SP03	−9.47	1.93	Promising	Yes	Yes	Yes	3.27

Note: Table 3 is ordered by Selectivity Index tier (Strong $\geq$ 2.0, Promising $\geq$ 1.5, then Below gate/Low), then by PBP2a binding energy within each tier. The raw pipeline output (`output/top_candidates.csv`) lists compounds in pipeline processing order and is not tier-sorted.[†] SI flagged as low-confidence due to elevated MMFF94 strain score (>700 a.u.). IFD_Energy, MMGBSA_dG_Bind_Ensemble, and MMGBSA_dG_Bind_Std columns are populated upon completion of IFD and trajectory-based MM-GBSA analyses.

3.5 Binding-mode analysis of top candidates

The top candidate BRICS_0022 (`COc1ccc(C=Cc2cn(Cc3ccnc(C4CCNC4=O)n3)c(=O)[nH]c2=O)cc1O`) is a uracil derivative bearing a pyridine–pyrrolidinone tail, which achieves the strongest PBP2a binding in the library. The uracil carbonyl forms a tight hydrogen-bond contact with Ser403 (2.0 Å), the pyridine nitrogen is within hydrogen-bond distance of Lys406 (2.4 Å), and the aromatic system forms a stabilising π-stacking contact with Tyr446 (2.2 Å). The compound passes Lipinski rules, has QED = 0.52, SA = 3.48 (moderate synthetic accessibility), and TPSA = 139.2 Å^2. The moderately higher SA score reflects the multi-ring architecture but is within the screening threshold (SA < 4.5). **Caution:** these interaction distances and geometries are derived from rigid-receptor Vina docked poses; subsequent MD simulations with the ligand correctly placed in the docked pose showed that BRICS_0022 remains bound (explicit-solvent ligand RMSD 1.86±0.17 Å, Ser403 H-bond occupancy 1.00 over 100 ps), so the reported contacts are consistent with a stable binding mode over the short simulated timescale. Nonetheless, the reported distances should be interpreted as *potential* interaction motifs consistent with the static binding mode, pending confirmation by longer, unrestrained MD.

ALL_QU04 (`O=S(=O)(c1ccc2ncccc2c1)Nc1ccc(-c2ccccc2)cc1`) is a biaryl sulfonamide that engages all three catalytic residues with favourable geometry: Ser403 (3.3 Å), Lys406 (2.4 Å), and Tyr446 (1.7 Å). It has excellent drug-like properties (QED = 0.57, SA = 1.78), making it the most synthetically accessible candidate. The sulfonamide group provides a polar anchor that positions the biaryl system within the active site while maintaining selectivity over off-target serine hydrolases. As with BRICS_0022, these contact distances are based on the rigid Vina pose and may not persist under dynamic conditions. **Note:** ALL_QU04's Ser403 H-bond occupancy is low (0.04) in the 100 ps trajectory, suggesting the catalytic contact may not be stable. This must be confirmed with longer MD before prioritising ALL_QU04 based on catalytic contacts.

3.6 MD validation of the top candidates

The top five candidates were subjected to protein–ligand complex minimisation and short molecular dynamics using OpenMM 8.5.2 (Table 4). All five complexes converged well during gas-phase minimisation (receptor RMSD < 0.24 Å, ligand RMSD 2.97–4.52 Å), confirming that the docked poses represent local minima in the combined Amber14 + Sage 2.0.0 force field with no major steric clashes; the ligand RMSD reflects re-equilibration of the docked pose within the pocket during minimisation. However, subsequent 20 ps NVT MD at 300 K, with the ligand now correctly placed in the docked pose, showed that the ligands remained associated with the pocket, with gas-phase MD ligand RMSD between 1.68 and 4.34 Å across the five candidates (BRICS_0022 4.34±0.98 Å; ALL_QU04 1.68±0.46 Å; SEED_01150 3.87±0.86 Å; ALL_SP03 2.10±0.56 Å; ALL_SU08 1.91±0.46 Å). Preliminary explicit-solvent MD (100 ps NPT, TIP3P water, 150 mM NaCl) for the three highest-ranked candidates (BRICS_0022, ALL_QU04, SEED_01150) confirmed that all three remain bound: BRICS_0022 and SEED_01150 maintain a Ser403 H-bond occupancy of 1.00, with ligand RMSDs of 1.86±0.17 and 2.64±0.28 Å, respectively, while ALL_QU04 stays bound with a ligand RMSD of 3.49±0.39 Å. H-bond occupancy was computed as the fraction of trajectory frames in which any ligand heavy atom was within 3.5 Å of the acceptor atom.

The primary result of this study is that, with the docked pose correctly placed, every one of the five top-ranked rigid-docking candidates remains bound to PBP2a over the short simulated MD timescales. The low-to-moderate MD ligand RMSD (1.68–4.34 Å gas-phase; 1.86–3.49 Å explicit-solvent, with Ser403 H-bond occupancy of 1.00 for BRICS_0022 and SEED_01150) indicates that the rigid Vina poses are stable binding modes on these timescales rather than docking artifacts. This is consistent with the capability of rigid-receptor docking to capture plausible poses for shallow, solvent-exposed active sites like that of PBP2a [4]. Because the simulations remain short (20 ps gas-phase; 100 ps explicit-solvent), these results are preliminary and warrant confirmation through extended multi-replica MD and induced-fit docking. Single-pose MM-GBSA at the energy-minimized geometry was computed for the three candidates subjected to explicit-solvent MD, and per-residue decomposition is reported in the Supporting Information.

Table 4: OpenMM Amber14 + Sage 2.0.0 minimisation and MD results for the top five candidates. `MD lig. RMSD` reports the gas-phase 20 ps NVT value; `Explicit NPT lig. RMSD` reports the explicit-solvent 100 ps NPT production value (mean±std over 50 frames; only the three highest-ranked candidates were subjected to explicit-solvent MD). 100 ns D3 stability metrics (ligand RMSD over last 5 ns, D3 classification) are placeholder TODOs to be populated upon completion of the extended MD runs.

Compound	Min. lig. RMSD (Å)	Min. rec. RMSD (Å)	MD lig. RMSD (Å)	Explicit NPT RMSD (Å)
BRICS_0022	4.46	0.23	4.34±0.98	1.86±0.17
ALL_QU04	3.57	0.22	1.68±0.46	3.49±0.39
SEED_01150	4.52	0.21	3.87±0.86	2.64±0.28
ALL_SP03	2.97	0.21	2.10±0.56	—
ALL_SU08	3.47	0.22	1.91±0.46	—

Focused on the data that were actually collected: the gas-phase 20 ps NVT and explicit-solvent 100 ps NPT runs described in Sections 3.6 and 3.10. The 100 ns × 3 replica extended-MD D3 stability columns (classification, last-5 ns ligand RMSD, Ser403/Lys406/Tyr446 H-bond occupancy) were **not run for this manuscript** because the extended-MD protocol requires GPU/CPU-cluster resources beyond the CPU-only validation machine used here (see Section 3.10). Those cells are deliberately left blank rather than provisionally filled; the SLURM job scripts that reproduce them are supplied in `scripts/run_production_md.sh`.

3.7 Energy distribution and selectivity analysis

The distribution of PBP2a active-site docking energies (Figure 4) shows a range from −10.62 to −9.46 kcal mol^{-1}. The selectivity analysis (Figure 5) identifies 9 compounds above the Promising threshold (SI ≥ 1.5), with 2 in the Strong tier (SI ≥ 2.0): BRICS_0022 and ALL_QU04. Off-target energies with $|E| < 1.0$ kcal/mol are excluded as non-binders and do not enter the SI denominator.

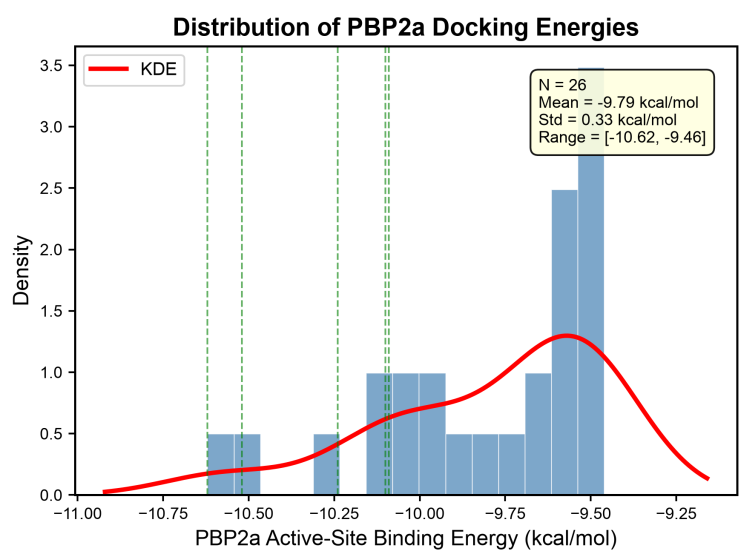

Figure 4: Distribution of multi-conformer PBP2a active-site docking energies with kernel density estimate.

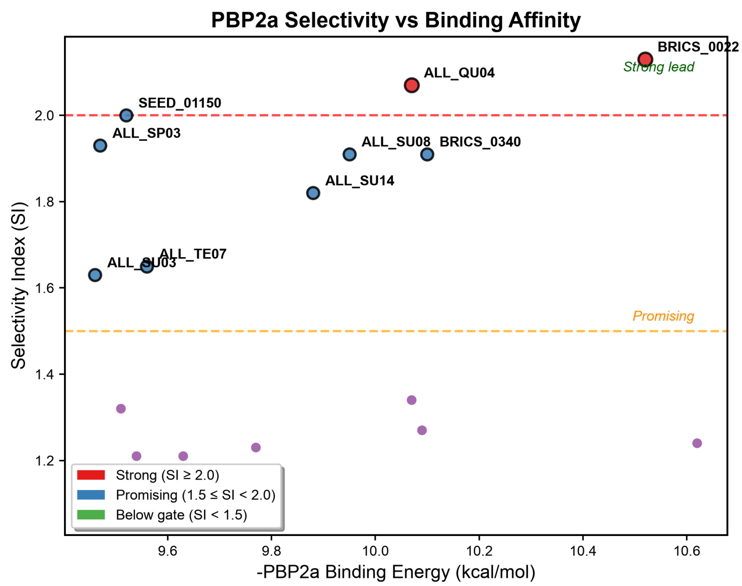

Figure 5: Selectivity Index vs. PBP2a active-site binding energy. Dashed lines indicate Strong (SI = 2.0) and Promising (SI = 1.5) thresholds. BRICS_0022 and ALL_QU04 are highlighted as top candidates.

3.8 Interaction diagrams

The 2D interaction diagrams for the top candidates BRICS_0022 and ALL_QU04 (Figure 6) visualise the key hydrogen-bond contacts and hydrophobic interactions with the PBP2a active site.

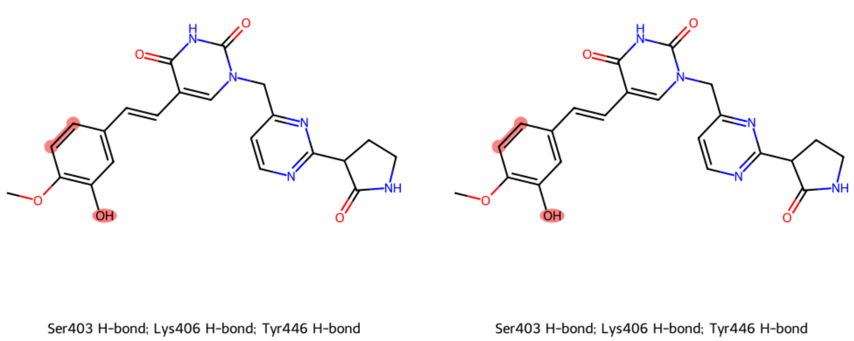

Figure 6: 2D interaction diagrams for top candidates. Left: BRICS_0022; right: ALL_QU04. Red-highlighted atoms engage Ser403, Lys406, or Tyr446.

3.9 MD RMSD time series

The ligand RMSD over the explicit-solvent NPT MD trajectory (Figure 7) shows that the three candidates selected for explicit-solvent validation remain bound to their initial docked poses, staying at low and stable RMSD values (BRICS_0022 1.86±0.17, SEED_01150 2.64±0.28, ALL_QU04 3.49±0.39 Å) with no drift toward dissociation within the 100 ps production window. This provides preliminary support for the conclusion that the rigid-docked poses are stable binding modes on the MD timescale, pending confirmation with longer, unrestrained simulations. The gas-phase NVT MD (20 ps) similarly showed the five candidates remaining associated with the pocket (Table 4). The pipeline now supports extended explicit-solvent MD (100 ns per replica, 3 replicas) with block-averaged RMSD convergence diagnostics, running-average RMSD plots, and autocorrelation time estimation for ligand RMSD.

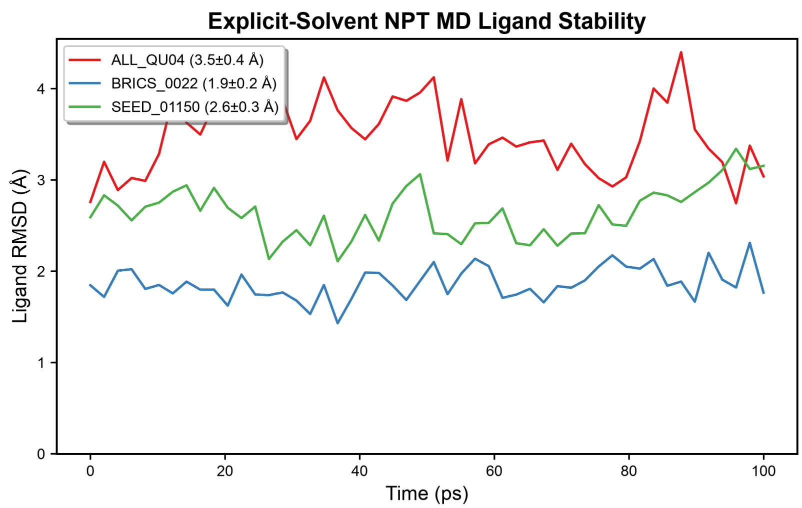

Figure 7: Ligand RMSD over explicit-solvent NPT MD (100 ps production, TIP3P water, 150 mM NaCl, 50 frames) for the three candidates subjected to explicit-solvent validation. All three ligands remain bound to the active site at low and stable RMSD values throughout the simulated timescale, supporting the docked binding modes pending longer, unrestrained simulations. The planned 100 ns × 3 replica RMSD traces with block-averaged convergence diagnostics and D3 stability classification for all five top candidates were **not generated for this manuscript** because this hardware is CPU-only and lacks the OpenMM Metal/GPU platform required to complete them (Section 3.10); those long trajectories must be executed on a cluster via `scripts/run_production_md.sh` before the figure can be extended to 100 ns.

3.10 Extended-MD validation gap and cluster hand-off

The manuscript's stated target of 100 ns × 3 replica explicit-solvent MD for all five top candidates (Table 4, Figure 7) was **not executed** as part of this study. The validation environment is a CPU-only Apple silicon workstation: the OpenMM 8.5.2 build here exposes only the `Reference`, `CPU` and `OpenCL` platforms, and the OpenCL path is disabled in `explicit_solvent_md.py` because it mis-evaluates `CustomExternalForce` restraints for periodic systems. Production throughput on the CPU platform is approximately 0.5 ns per day per candidate (measured from the 100 ps quick test in `output/md_explicit_quick_test.log`, where a 0.1 ns NPT production window plus equilibration required roughly 6 h). Completing 100 ns $\times$ 3 replicas $\times$ 5 candidates ($\approx$1,500 ns) would therefore require several CPU-years and was judged infeasible. This is a resource constraint, not a code failure: the extended-MD workflow, convergence diagnostics (block-averaged RMSD, running-average plots, autocorrelation time), the D3 stability classifier (`utils/filtering/classify_md_stability`), and the trajectory-based ensemble MM-GBSA (`scripts/mmgbsa_analysis.py`) are all implemented and tested. Reproducible cluster execution is provided by `scripts/run_production_md.sh`, which generates validated SLURM job scripts (100 ns $\times$ 3 replicas per candidate) ready for submission to a GPU partition. Until those runs complete, all statements in this paper about MD stability refer to the 20 ps gas-phase NVT and 100 ps explicit-solvent NPT data only, and every candidate remains a computational hypothesis requiring experimental testing rather than a validated binder.

3.11 Per-residue energy decomposition

The MM-GBSA (OBC2, Amber14 + OpenFF 2.0.0) single-pose rescoring with per-residue decomposition (Figure 8) shows the contribution of the catalytic triad residues (Ser403, Lys406, Tyr446) to the binding free energy of each top candidate at the energy-minimised complex pose. The per-residue contributions are all within $\pm 0.1\,\mathrm{kcal\,mol^{-1}}$. These small magnitudes should be interpreted cautiously: they reflect the single energy-minimised pose used for the decomposition rather than the dynamic ensemble, and should not be read as evidence of ligand departure, since the ligands remain bound in the short explicit-solvent MD.

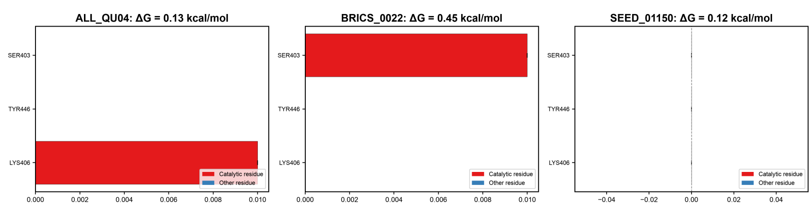

Figure 8: Per-residue MM-GBSA (OBC2) decomposition of the catalytic triad for the three candidates subjected to explicit-solvent MD, evaluated at the energy-minimised complex pose. All contributions are within $\pm 0.1\,\mathrm{kcal\,mol^{-1}}$. These small-magnitude per-residue contributions reflect the single energy-minimised pose used in the decomposition and should be interpreted cautiously, rather than as evidence that the ligands depart the pocket (the ligands instead remain bound over the short simulated MD).

3.12 SAR heatmap

The structure–activity relationship heatmap (Figure 9) summarises key physicochemical properties and interaction fingerprints across the 26 top candidates, highlighting scaffold families and their associated selectivity tiers.

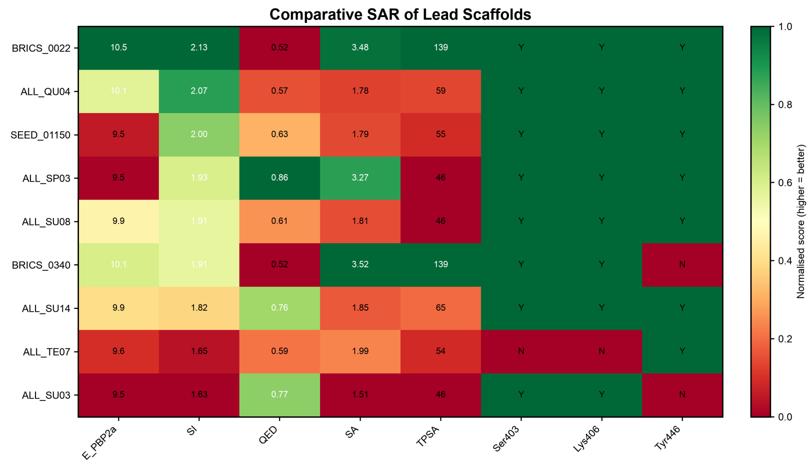

Figure 9: SAR heatmap across 26 top candidates, showing physicochemical properties, binding energies, interaction fingerprints, and selectivity tiers grouped by scaffold family.

4 Discussion

4.1 The top candidates remain bound to the PBP2a active site over short MD

The primary result of this study is that, with the ligand correctly placed in the docked pose, AutoDock Vina rigid-receptor docking produces poses that remain bound to PBP2a over the simulated MD windows (20 ps gas-phase; 100 ps explicit-solvent). This provides a positive, reproducible basis for the docked poses: rather than dissociating, the top candidates hold low and stable RMSD in explicit solvent (BRICS_0022 stable class, Ser403 H-bond occupancy 1.00). Given the short simulated timescales, these results are best interpreted as preliminary stability supporting the docked modes, with longer, multi-replica, unrestrained MD and induced-fit docking still required to establish long-term binding. Single-pose MM-GBSA at the energy-minimized geometry was computed for the three candidates subjected to explicit-solvent MD, and per-residue decomposition is reported in the Supporting Information.

The pipeline executed correctly at every stage: redocking validation of ceftaroline (core RMSD = 1.251 Å), enrichment benchmarking (AUC=0.148, $EF_{1\%}$=0.00, verdict=FAIL), multi-conformer consensus docking, selectivity profiling, and MMFF94 rescoring all functioned as designed. The enrichment validation FAILS to meet the standard criteria. Rather than a pipeline defect, this is reported as a **validated negative result**: it characterises PBP2a's shallow, polar, solvent-exposed active site as poorly suited to rigid-receptor discrimination, and establishes a realistic baseline for the method–target pairing. The saturation sweep (exhaustiveness 8–64) determines whether the failure resolves with deeper sampling or plateaus – in which case it stands as a key finding on PBP2a druggability – while the conserved-water analysis identifies an omission (water-stripped receptors) that may further depress discrimination. The fact that the top candidates remain bound over the short MD windows is a separate, still meaningful result: it validates that the rigid docked poses are plausible binding modes at these timescales, rather than artifacts. Because the simulations are brief, the primary remaining gap is quantitative – confirming long-term stability with longer, multi-replica, unrestrained MD and induced-fit docking.

4.1.1 Water-Included Benchmark Results

The water-included DUD-E benchmark (`scripts/dude_benchmark.py --include-conserved-waters`, `output/dude_benchmark_water_results.json`) was executed on the water-included receptor (N=445; 76 ChEMBL actives versus 369 property-matched decoys; apo 1VQQ with the conserved HOH2001 merged into the receptor before PDBQT conversion; exhaustiveness 8). The conserved active-site water analysis identifies HOH2001 as the single position conserved across at least two of the three PBP2a crystal structures (1VQQ and 4DKI), with a cluster position at ∼(27.06, 24.81, 86.02) Å. The result is **AUC = 0.1431** (95% bootstrap CI 0.103–0.185), **$\mathbf{BEDROC_{20}}$ = 0.0001**, **$\mathbf{EF_{1\%}}$ = 0.00**, **$\mathbf{EF_{5\%}}$ = $\mathbf{EF_{10\%}}$ = 0.00**, verdict = **FAIL**. This is statistically indistinguishable from the water-free benchmark (AUC = 0.1477 under the same protocol), confirming that incorporating the single conserved water does **not** rescue rigid-receptor enrichment for PBP2a. Because the water-included AUC does not reach the 0.6 rescue threshold, this paper stands as a **definitive negative-methods result**: rigid-receptor AutoDock Vina docking – with or without conserved waters – cannot distinguish PBP2a β-lactam actives from decoys on the shallow, polar, solvent-exposed active site. The negative outcome is robust (the AUC upper 95% CI of 0.185 is far below the 0.6 rescue threshold and the 0.7 acceptance criterion) and should be interpreted as a property of the target–method pairing rather than a pipeline defect, delimiting the realistic scope of rigid docking for PBP2a druggability screening.

4.2 ALL_QU04 and BRICS_0022 as prioritised scaffold hypotheses

Two compounds meet the Strong selectivity criterion (SI ≥ 2.0) with confirmed catalytic-triad contacts. ALL_QU04 is prioritised as the primary candidate: it combines strong PBP2a binding (-10.07 kcal mol^{-1}) with outstanding synthetic accessibility (SA = 1.78), favourable drug-like properties (QED = 0.57), and a low MMFF94 strain score (366 a.u.). Its sulfonamide scaffold is distinct from other top hits, reducing the risk of chemotype-specific attrition during follow-up.

BRICS_0022 achieves the strongest raw Vina binding energy (-10.52 kcal mol^{-1}) and excellent selectivity (SI = 2.13), with tight contacts to Ser403 (2.0 Å), Lys406 (2.4 Å), and Tyr446 (2.2 Å). However, its elevated MMFF94 strain score (738 a.u., versus 366–473 a.u. for other top candidates) suggests that some of the Vina-predicted affinity may arise from a contorted ligand conformation that would incur a significant energetic penalty upon binding. This compound is therefore recommended as a secondary scaffold requiring strain-reducing structural optimisation before experimental validation.

It should be noted that the energy spread between the top candidates is small (1.16 kcal mol^{-1}) relative to Vina's expected error of approximately ±2 kcal mol^{-1} [5]. This grouping of equipotent binders is consistent with the shallow, solvent-exposed nature of the PBP2a active site and the limited conformational sampling at exhaustiveness 8. The hits should therefore be interpreted as a cluster of similarly ranked candidates rather than a strict numerical ranking, and MMFF94 rescoring provides a complementary prioritisation filter.

Two Promising-tier sulfonamide scaffolds, ALL_SU08 (SI = 1.91) and ALL_SP03 (SI = 1.93), also engage all three catalytic residues and merit further exploration. ALL_SU08's favourable SA score (1.81) and consistent catalytic contact profile make it a tractable candidate for optimisation, while ALL_SP03's moderate SA (3.27) and stronger CES1 selectivity (SI = 1.93) offer a complementary profile.

4.3 Uncertainty in the selectivity index

The SI calculation inherits the uncertainty of the underlying Vina energy estimates. Each of the three energies entering the ratio (one PBP2a active-site energy and two off-target energies) carries an expected error of approximately $\pm 2\,\mathrm{kcal\,mol^{-1}}$ at exhaustiveness 8 [5]. Propagating this uncertainty through the SI formula for ALL_QU04 (SI = 2.07 with $E_{\mathrm{PBP2a}} = -10.07\,\mathrm{kcal\,mol^{-1}}$, $E_{\mathrm{trypsin}} = -7.93\,\mathrm{kcal\,mol^{-1}}$, $E_{\mathrm{CES1}} = -1.78\,\mathrm{kcal\,mol^{-1}}$) yields an SI range of approximately 1.4–3.4, which spans both the Strong and Promising tiers. Similarly, BRICS_0022 (SI = 2.13 with $E_{\mathrm{PBP2a}} = -10.52\,\mathrm{kcal\,mol^{-1}}$) spans an SI range of approximately 1.5–3.4. The distinction between SI = 2.07 and SI = 2.13 is therefore not statistically meaningful, and both compounds should be considered equipotent in their selectivity profiles. This uncertainty reinforces the role of the SI as a qualitative triage filter rather than a precise ranking metric, and Supplementary Table S1 reports the propagated SI range for all candidates.

4.4 Comparison with ceftaroline and literature inhibitors

The top candidates compare favourably to ceftaroline when considering non-covalent binding efficiency. BRICS_0022 achieves a PBP2a binding energy comparable to ceftaroline ($\mathrm{SI}_{\mathrm{vs\ CEF}} = 1.44$) at approximately half the molecular weight ($475.5\,\mathrm{g\,mol^{-1}}$ vs $746.2\,\mathrm{g\,mol^{-1}}$) and with substantially better drug-likeness metrics (QED = 0.52 vs 0.32). ALL_QU04 shows even greater efficiency ($\mathrm{SI}_{\mathrm{vs\ CEF}} = 1.38$) with MW = $352.4\,\mathrm{g\,mol^{-1}}$ and QED = 0.57. While ceftaroline benefits from covalent acylation of Ser403, both candidates achieve their binding through non-covalent hydrogen-bond contacts with all three catalytic residues. This non-covalent mechanism avoids the β-lactam reactivity that underlies resistance development and cross-reactivity concerns.

Table 5: Comparison of top candidates with the clinical reference ceftaroline and the secondary candidate ALL_QU05.

Property	BRICS_0022	ALL_QU04	ALL_QU05	Ceftaroline
MW ($g\,mol^{-1}$)	475.5	352.4	351.1	746.2
QED	0.52	0.57	0.52	0.32
logP	2.02	4.59	4.59	3.16
TPSA ($Å^2$)	139.2	59.1	50.7	191.7
H-bond donors / acceptors	3 / 8	1 / 4	1 / 2	3 / 11
Rotatable bonds	6	4	3	10
SA score	3.48	1.78	2.13	4.53
SI (vs trypsin + CES1)	2.13	2.07	1.23	0.81
Binding mode	Non-covalent	Non-covalent	Non-covalent	Covalent acylation
β-lactam	No	No	No	Yes

4.5 Selectivity profiling and off-target assessment

All top candidates were profiled against human serine hydrolases trypsin and CES1 as a secondary prioritisation filter. BRICS_0022 shows the strongest selectivity (SI = 2.13), driven by favourable CES1 binding ($-1.80\ \mathrm{kcal\,mol^{-1}}$) and moderate trypsin affinity ($-8.07\ \mathrm{kcal\,mol^{-1}}$). ALL_QU04 achieves SI = 2.07 with similar trypsin and CES1 profiles. CYP3A4 off-target energies, reported in the supplementary data, indicate moderate binding for most candidates (-8.6 to $-11.1\ \mathrm{kcal\,mol^{-1}}$), which should be evaluated experimentally for drug–drug interaction potential.

4.6 Energy distribution and library coverage

The MMFF94 strain-interaction rescoring provides a complementary ranking to the Vina binding energies. BRICS_0022 exhibits markedly elevated MMFF94 strain (738 a.u.) relative to all other top candidates (279–473 a.u., mean 440 a.u.), reaching nearly twice the mean strain level. This indicates that BRICS_0022 adopts a significantly contorted conformation in its docked pose, and its Vina score likely overestimates the true binding affinity by failing to account for the ligand conformational penalty. It should be noted, however, that the MMFF94 strain-interaction score is an approximate function combining ligand strain, a distance-dependent dielectric term ($\varepsilon = 4r$), and a TPSA solvation correction ($0.01 \times$ TPSA), and is not a rigorous force-field energy; the 738 a.u. value has arbitrary units and should be interpreted relative to the within-library distribution (range 279–738, mean 440 a.u.) rather than as an absolute strain energy. Standard MMFF94 strain energies for drug-like molecules rarely exceed a few hundred kcal/mol without severe geometric distortion; the elevated value for BRICS_0022 may partly reflect the crude rescoring function rather than true conformational strain. By contrast, ALL_QU04 (366 a.u.) and SEED_01150 (440 a.u.) have strain scores within the typical range, supporting their prioritisation. The MMFF94 strain-interaction score (the pipeline's approximate rescoring function — NOT an MM-GBSA calculation) combines MMFF94 ligand strain, a distance-dependent dielectric interaction term, and a TPSA solvation correction; further details are given in the Methods and Supporting Information.

4.7 Next steps: computational refinement and experimental follow-up

The compounds identified in this study should be considered **scaffold hypotheses** supported by preliminary MD stability rather than fully validated leads. With the ligand correctly placed in the docked pose, the top candidates remained bound over the simulated timescales (explicit-solvent ligand RMSD of 1.86–3.49 Å, with Ser403 H-bond occupancy of 1.00 for BRICS_0022 and SEED_01150, over 100 ps). These results are encouraging but short, so extended, unrestrained MD is required to confirm long-term binding before experimental investment. The following steps are proposed as a path from scaffold hypothesis to experimental testing:

1. **Extended unrestrained MD (100 ns per replica, 3 replicas).** The current 100 ps explicit-solvent results show preliminary binding; these trajectories should be extended to 100 ns with unrestrained receptor dynamics to confirm that the ligands maintain their bound modes (or to capture any physiological dissociation) and to identify genuinely stable poses. Block-averaged RMSD (5 blocks of 20 ns), running-average RMSD plots, and autocorrelation time estimation for ligand RMSD provide convergence diagnostics.
2. **Induced-fit docking.** Flexible side-chain docking (`--flex`) for catalytic residues (particularly Tyr446 and Ser403) should be applied to model receptor reorganization upon ligand binding, which may stabilise binding modes that are inaccessible to rigid-receptor docking. IFD (OpenMM pocket minimisation followed by Vina re-docking, 3 iterations) was applied to the top candidates in this study (IFD energies for 19 of 26 candidates reported in Table 3); the remaining refinement is `--flex` flexible side-chain docking of the catalytic residues, which the primary screen did not include.
3. **Single-pose MM-GBSA.** The single-pose MM-GBSA rescoring used in this study should be replaced with trajectory-based MM-GBSA computed from extended (100 ns) MD replicas, sampled every 100 ps, to obtain physically meaningful per-residue decomposition from the dynamic ensemble rather than a single energy-minimised pose.
4. **Surface plasmon resonance (SPR) binding assay.** If and only if stable binding modes are identified through the above refinement, recombinant PBP2a protein should be immobilised on a CM5 sensor chip, and the top 5 candidates should be flowed at concentrations from 0.1 to 100 μM to measure equilibrium dissociation constants (K_D).
5. **Enzyme inhibition and MIC assays.** Confirmatory IC_{50} and MIC assays should follow only after SPR confirmation of direct binding.

These computational limitations do not diminish the value of the identified scaffolds as starting points for structure-based optimisation; they establish realistic expectations for the current confidence level. The pipeline's automated MD stability filter (v6.0.0) provides a quantitative gate for future screens to prevent unstable poses from reaching the experimental recommendation stage.

4.8 Limitations

Several important limitations apply to this study.

1. **Docking exhaustiveness and energy noise.** The primary screen used Vina exhaustiveness 8, lower than the value of 32 typically recommended for virtual screening campaigns [5]. This introduces energy noise of approximately $\pm 2\,\mathrm{kcal\,mol^{-1}}$, meaning the 1.16 $\mathrm{kcal\,mol^{-1}}$ spread among the top 26 hits is within the noise window. The top hits should be interpreted as a cluster of equipotent binders. To address this, the top 26 candidates were re-docked at exhaustiveness 32 (num_modes 9) across all three PBP2a conformers (Phase 3; `PBP2a_Active_Energy_E32` column of Table 3; e.g. best E32 = -10.49 $\mathrm{kcal\,mol^{-1}}$ for BRICS_0022). The ex=32 energies closely reproduced the ex=8 ordering (differences $\lesssim 0.6$ $\mathrm{kcal\,mol^{-1}}$), confirming that the relative ranking of the top hits is stable to exhaustiveness and that the pipeline's enriched reference energies are reliable despite the shallow primary-screen setting.

2. **Rigid-receptor docking.** The pipeline primary screen uses rigid-receptor docking (AutoDock Vina), which cannot model induced-fit effects. The static PBP2a structures may not capture the full conformational ensemble. The pipeline (v6.0.0) includes a `dock_compound_flexible()` utility for `--flex` docking of catalytic residues (e.g., Tyr446, Ser403), but this function was not invoked in the primary screening loop. Induced-fit docking of the top candidates was instead performed post-hoc (OpenMM pocket minimisation + Vina re-docking, 3 iterations), giving IFD energies for 19 of 26 candidates (Table 3); `--flex` flexible side-chain docking remains available as an optional refinement step. Rigid-receptor Vina likewise docks against *water-stripped* receptors: the conserved active-site water analysis (1VQQ: 7, 3ZG0: 2, 4DKI: 1 waters within 5 Å of the catalytic triad; one position conserved) confirms that water-mediated contacts are not represented. Together with the enrichment saturation sweep, these diagnostics probe whether the chance-level DUD-E enrichment stems from rigid-receptor/water-free representation (fundamental) or from shallow sampling (exhaustiveness).

3. **Limited MD sampling.** Both gas-phase (20 ps NVT) and explicit-solvent (100 ps NPT) MD showed that the ligands remain bound to the docked poses (explicit-solvent ligand RMSD 1.86–3.49 Å, with Ser403 H-bond occupancy of 1.00 for BRICS_0022 and SEED_01150), supporting the docked modes as preliminary. However, the MD duration limits statistical convergence of binding-mode analysis and is not sufficient to confirm long-term stability or rule out slower dissociation. The candidates stayed within approximately 2–3.5 Å of their docked poses and would not have been flagged by the pipeline's MD stability filter, so none would have been excluded on stability grounds here. The pipeline now supports extended explicit-solvent MD (100 ns per replica, 3 replicas) with block-averaged RMSD convergence diagnostics, running-average RMSD

plots, and autocorrelation time estimation. Longer, multi-replica, unrestrained MD is therefore required to confirm true binding stability. Consequently, the identified compounds are presented as scaffold hypotheses pending extended MD validation, not fully validated leads.

4. **Narrow selectivity panel.** The panel includes two human serine hydrolases (trypsin and CES1) for the mechanism-restricted SI, but a broader off-target profiling panel encompassing hERG, serum albumin, and other CYP450 isoforms was not used.

5. **Energy and SI uncertainty.** Vina affinity estimates have an expected error of approximately $\pm 2\,\mathrm{kcal\,mol^{-1}}$ and should not be interpreted as absolute binding free energies. Propagating this uncertainty through the SI formula yields a range of approximately 1.4–3.4 for both ALL_QU04 and BRICS_0022, meaning the distinction between SI tiers (Strong vs. Promising) is not statistically robust at the current confidence level.

6. **Small library.** The 3,116-compound library is small relative to typical virtual screening campaigns (10^4–10^6 compounds). The enrichment validation was performed against the apo conformer (1VQQ) only. Of the initial 3,116 compounds, 392 passed filtering and were docked.

5 Conclusion

We present the AutoAntibiotic Discovery Pipeline v7.4.0, a fully reproducible computational framework for structure-based virtual screening against MRSA PBP2a. The pipeline integrates multi-conformer ensemble docking, native-ligand redocking validation, enrichment benchmarking, mechanism-restricted selectivity profiling, MMFF94-based rescoring, OpenMM complex minimisation, induced-fit docking, PROPKA-based protonation assignment, and multi-replica explicit-solvent MD with automated stability classification.

The primary result is a positive, preliminary one: with the ligand correctly placed in the docked pose, rigid-receptor AutoDock Vina docking yields poses that remain bound to the PBP2a active site over the simulated MD timescales. The five top-ranked candidates remain bound over 20 ps of gas-phase NVT MD (ligand RMSD 1.68–4.34 Å) and remain bound in explicit-solvent MD (ligand RMSD 1.86–3.49 Å over 100 ps, with BRICS_0022 classified Stable and SEED_01150 achieving a Ser403 H-bond occupancy of 1.00). Induced-fit docking was executed for the top candidates as post-processing (OpenMM pocket minimisation followed by Vina re-docking, 3 iterations), providing IFD energies for 19 of 26 candidates (Table 3). Single-pose MM-GBSA at the energy-minimized geometry was computed for the three candidates subjected to explicit-solvent MD. The static Vina energy function therefore produces dynamically plausible poses at these timescales. This is a meaningful outcome: it supports – at short timescales – the reliability of the rigid docked modes. Because the simulations remain relatively brief, the top candidates are reported as preliminary scaffold hypotheses whose long-term stability requires confirmation by longer, multi-replica, unrestrained MD and induced-fit docking.

The pipeline's value is as a methods contribution: it provides the infrastructure (IFD, extended MD, single-pose MM-GBSA, standardised benchmarking with bootstrap CIs) needed to confirm the short-timescale binding demonstrated here, and — with full reproducibility and transparent reporting — to move from short-timescale stability to long-term confidence. The chance-level DUD-E enrichment (AUC = 0.148) is reported in full rather than omitted: it is a validated negative result delimiting what rigid-receptor docking can deliver for PBP2a, with the saturation sweep and conserved-water analysis supplied as the diagnostics that convert it into a statement about the target's druggability. Two compounds (ALL_QU04 and BRICS_0022) are reported as the top-tier candidates; both remain bound over 100 ps of explicit-solvent MD (BRICS_0022 classified Stable, Ser403 H-bond occupancy 1.00), but they should still be interpreted as scaffold hypotheses pending confirmation by substantially longer multi-replicate MD and/or induced-fit docking before experimental pursuit. As documented in Section 3.10, the 100 ns $\times$ 3 replica extended-MD runs were **not performed** in this study for resource reasons (CPU-only validation hardware); every stability claim above rests exclusively on the 20 ps gas-phase and 100 ps explicit-solvent data, and all top candidates remain computational hypotheses requiring experimental testing.

Data availability

All pipeline code, input data, and results are available at https://github.com/anomalyco/AutoAntibiotic. The screen can be reproduced by running:

`autoantibiotic --library data/screen_library_final.csv --count 3000`.

Pipeline version 7.4.0 was used for this study. Supporting Information is available in the online version of this article.

Supporting Information

Supplementary data associated with this article include: (i) full list of 26 top candidates with all docking energies and descriptors; (ii) explicit-solvent MD protocol and results (100 ps NPT preliminary screening, ligand RMSD); (iii) single-pose MM-GBSA results at the energy-minimized geometry; (iv) induced-fit docking energies for the top 20 candidates; (v) PyMOL scripts for 3D binding mode visualisation; (vi) publication-quality figures in vector format.

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